

Using Google Colab for AI Model Training

1. Open a Colab notebook

- Go to <https://colab.research.google.com> and start a New Notebook.
- Set Runtime → Change runtime type → Hardware accelerator → GPU (T4 on the free tier).

Free-tier GPU access is **not guaranteed** every session. You may occasionally get a CPU-only runtime or run out of GPU quota for a while. **Start early**, keep individual runs reasonably short, and **save your progress often** (see Section 6).

2. Mount your Google Drive

Your code and the MicroSeg dataset both need to be accessible from Colab. The simplest way to do both is to keep them in your Google Drive and mount it once per session:

```
from google.colab import drive
drive.mount('/content/drive')
```

The dataset is large, so **it should not be committed to GitHub**. Download it into your Google Drive, mount it (see above), then point your data path at it.

Keep a copy of your project folder in your Google Drive (you can use Google Drive for Desktop, or simply upload the folder at drive.google.com), and run it directly from there.

```
%cd /content/drive/MyDrive/<path-to-your-project-folder>
```

Important Note: Colab is for training and testing your models, not for development. Your GitHub repository is the official record of your code and commit history. Keep committing your real changes there.

3. Install dependencies

Install whatever your code imports, the same way you would locally:

```
!pip install torch torchvision numpy matplotlib pillow
```

4. Run training and evaluation

Once your Drive is mounted and dependencies are installed, run your pipeline exactly as you would locally, using `run.py` in the appropriate mode. If you modified the starter structure, adjust the commands below to match your own entry point:

```
!python run.py --mode train --data_dir /content/drive/MyDrive/<path-to-microseg> --out_dir results/
!python run.py --mode eval --checkpoint results/best_model.pt --data_dir /content/drive/MyDrive/<path-to-microseg>
```

Important: Colab's usually stores output files in a temporary location that is wiped when your runtime disconnects or times out. To preserve your files, save your results, outputs, model checkpoints to a Google Drive location (mounted in Section 2) instead of relying on Colab's local disk, to avoid loss of progress. For example, point `--out_dir` at a folder inside MyDrive, e.g. `/content/drive/MyDrive/<results_dir>` instead of simply `<results_dir>/`.